



Artificial intelligence powered crime scene analysis service^{☆,☆☆}

Vanita Kshirsagar^{a,*}, Nishant Pachpor^b, Shubhangi Suryawanshi^a, Tanvi Chavan^a,
Navya J. Nair^a, Purvesh Agrawal^a, Tejas Shahane^a

^a Dr. D. Y. Patil Institute of Technology, Pimpri, Pune

^b International Institute of Management Science, Pune



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ABSTRACT

Crime remains a major concern in modern culture. As a result, emphasizing prevention and ensuring prompt justice delivery are critical. In criminal investigations and law enforcement, forensic science plays a critical role. However, typical approaches in this field frequently rely on physical procedures, which are inefficient and likely to increase human mistakes, hampering the timely administration of justice. To tackle these issues, artificial intelligence skills were integrated into investigation processes. The suggested novel web application aims to help forensic investigators streamline crime scene investigations by automating important portions of the procedure.

The system has three main functions:

- fingerprint reconstruction,
- weapon detection, and
- human activity recognition.

Its easy-to-use interface allows officials to upload images or videos from the crime scene in order to assist in reconstructions and detections. The fingerprint reconstruction model built using autoencoders outputs better partially covered fingerprint images with a validation loss of 0.0477 and a step loss of 0.0487, which helps detect the persons responsible for the event. Furthermore, the model for object detection, YOLO NAS, plays an important role in recognizing weapons that might be present at the scene, with an mAP of 77.8 %, while human activity detection techniques such as VGG16 have a total accuracy of 98.21 %.

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* Corresponding author.

E-mail addresses: vanita.kshirsagar@dypvp.edu.in, vanita.kshirsagar@gmail.com (V. Kshirsagar).

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Background

Forensic science, a multidisciplinary discipline that addresses biology, chemistry, physics, and law, plays a crucial role in solving complex criminal investigations by providing scientific analysis and expert witness testimony on physical evidence [1]. In the heart of this multidimensional world, crime scene investigations are the critical initial step where evidence is systematically taken, protected, and meticulously analyzed. This process is required to rebuild events, to define critical time frames, and to track the persons responsible for criminal actions. These studies need to be both effective and accurate because they form the basis of the search for justice. The main focus of the work is to design a system that is easy to use so that forensic investigators can perform deep crime scene investigations in a short time frame and with high efficiency. Using deep learning techniques [2], the goal is to produce a versatile tool that streamlines everyday operations while improving forensics experts' analytical capabilities. The proposed system is designed to automate a large number of key tasks, such as fingerprint generation, detection of weapons and unknown objects [1], and activity recognition of humans at crime scenes.

The advantages of integrating those technologies into a single, easily navigated system that will enable data integration and improved forensic examination. In addition, to present the possibility of integrating IoT to implement the development of a portable device that will be able to take photos and videos and perform the anticipated functionalities in a rapid, automatic way. The paper offers insights for automating those critical domains, with the goal of streamlining investigative proceedings and, ultimately, helping to deploy a more efficient, effective criminal justice system. The study has a great impact on forensic science and criminal investigation. Following is a set of contributions to this work. The suggested online application combines three unique functionalities into a cohesive platform, adding innovation to the current literature. The system standardizes and integrates key aspects of crime scene interrogation, saving time and resources for forensic work and enabling law enforcement to bring justice to resolution more quickly.

AI technology improves the precision and dependability of forensic analysis, leading to more effective evidence collection and investigative methods. Diverse datasets enable effective model training for various test situations, ensuring relevance in real-world scenarios. The engine of design's plan is a system of deep learning algorithms, consisting of autoencoders for fingerprint synthesis and state-of-the-art models for object detection (YOLOv5 [1,2] and YOLO-NAS [3,4]) to detect weapons and human activity recognition (HAR) using VGG-16 [5,6]. These sophisticated algorithms allow us to glean meaningful information from noisy data with high efficiency, enabling accurate and rapid forensic analysis.

Researchers have achieved great strides in specific functionalities such as weapon detection, human activity recognition, and fingerprint reconstruction, according to a review of earlier work in this area. For example, in "Using Generative Adversarial Networks Synthesis and Reconstruction of Fingerprints" [7,8], researchers examine how to synthesize and reconstruct fingerprints utilizing the StyleGAN2 architecture as the fingerprint generator. The Minutiae-to-Vec encoder, which they presented, outperforms current methods by providing an accuracy of 69.20 % for real fingerprint datasets, 71.43 % for synthetic datasets, and 83.72 % for the combination of both types of data. It encodes minutiae into latent vectors for fingerprint reconstruction. The importance of automated threat item recognition in X-ray pictures for security is discussed by Jola Koci et al. [9], who emphasize the use of several detection models and achieve the best accuracy of 87.58 % with Faster R-CNN and ResNet101 [10]. In a cross-dataset evaluation, [11] and [12] offer a user-independent deep learning-based method and convolutional neural network used for online human activity classification and statistical features. This method outperforms existing approaches on the UCI HAR and WISDM datasets while retaining user and platform independence. A novel method for producing artificial fingerprint images is presented by R. Cappelli et al. [13], which can be applied to the construction of large datasets for the purpose of evaluating and refining identification algorithms [14] provides a framework that is 19 times quicker than the quicker R-CNN model [15] and is based on YOLOv5s and area of interest to improve weapon identification by preserving speed, achieving high recall rates, and precision on video datasets. There aren't any customized guns in the dataset that don't just look like regular pistols. To recognize firearms in public spaces, convolutional neural networks are utilized [16]. The significance of computer vision and machine learning techniques in boosting the effectiveness of surveillance systems, which aim for a mean average precision (mAP) of 70 %, is highlighted. This focus on classification and detection performance serves as the basis for evaluation. However, the homogeneous dataset has its limitations [17,18]. A framework mentioned in the literature is 19 times faster than the already quick R-CNN model, leveraging YOLOv5s and area of interest to enhance weapon identification while maintaining speed and achieving high recall rates and precision on video datasets. Notably, the dataset does not include modified firearms that differ from the standard pistol design. In order to detect weapons in public spaces, convolutional neural networks are employed in [16]. It highlights the importance of computer vision and machine learning techniques in surveillance systems that are effective, with performance in detection and classification acting as the foundation for assessment and reaching a 70 % mean average precision (mAP). The homogenous dataset presents limitations.

Yuhan et al. [19] This work presents a new approach for identifying changes in human activity detection based on the mean of the geometric moving average of the Martingale sequence [20,21]. In the present study, an innovative model for human activity recognition (HAR) called DeSepTr is suggested. DeSepTr outperforms previous models on several HAR tasks and could help develop more dependable and widely applicable HAR models [22]. The study explores the use of conditional generative adversarial networks (GANs) to rebuild realistic fingerprint patterns from minutiae maps. Minutiae maps are grayscale pictures that show minutiae's orientations and locations [23]. A novel fingerprint enhancement technique based on decimation-free directional filter banks (DFBs) is presented in this work. The suggested technique substantially improves fingerprint photographs by maintaining ridge structures and enhancing ridge clarity [24]. Minute details are what set fingerprint representation apart; however, it was previously believed that these details weren't enough to recreate the original fingerprint image [25]. Using object recognition algorithms like YOLO, SSD, and Faster R-CNN, Mostafa Azizi and Mohammed Boukabous propose a DL-based monitoring system for identifying crimes. YOLOv5, with its excellent accuracy and speed, is the best algorithm for real-time item recognition in security and surveillance

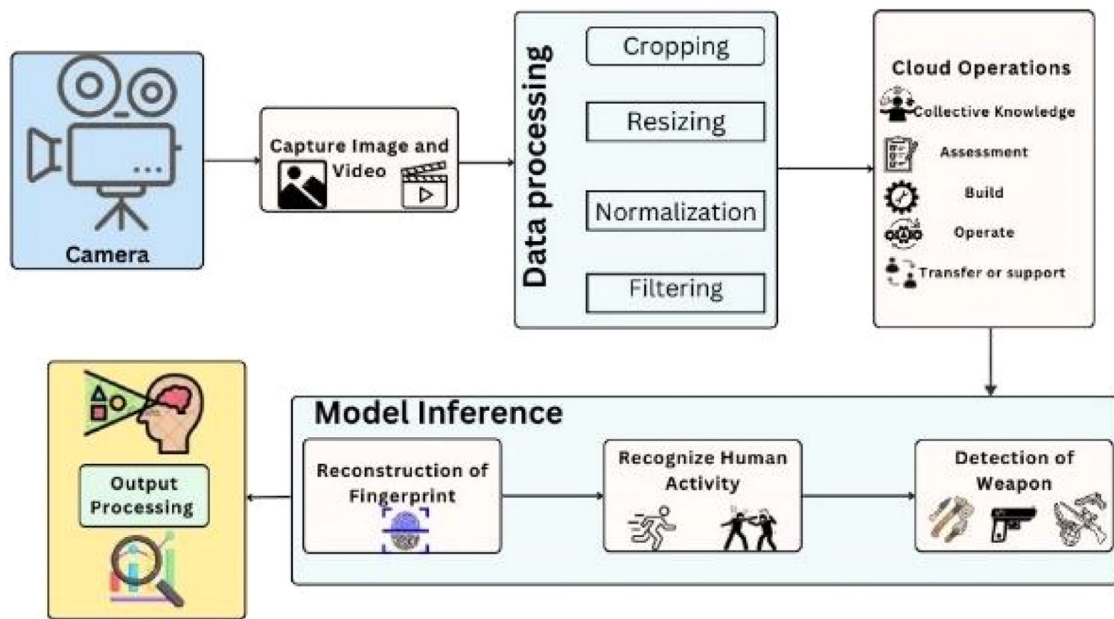


Fig. 1. Proposed multimodal crime scene analysis architecture.

applications, according to their comparative analysis [26]. The paper compares SSD and Faster R-CNN, where SSD gives faster real-time detection, but Faster R-CNN offers higher precision, demonstrating the compromise between speed and accuracy in weapon identification algorithms.

After reading these studies, I conclude that using AI technology to investigate crime scenes has a lot of possibilities. There isn't a single system in research that can automate investigation procedures. A gadget that is capable of handling several functions at once and producing timely and accurate results is required. This review provides us with a good foundation for upcoming improvements.

Method details

The system's design is shown graphically in Fig. 1, which also shows the thorough method of crime scene investigation. It employs a smooth approach towards the capturing of images and videos at the crime scene, beginning with the placement of a hardware device that possesses cameras and a user-friendly interface. Thereafter, data is transferred safely to cloud storage using Google Drive for easy access and storage. Data preprocessing in the cloud environment is carried out to prepare the data for model inference, capitalizing on the availability of processing resources in the cloud for optimal computational efficiency.

Model inference comprises particular algorithms for different systems, including human activity recognition (HAR), weapon detection, and fingerprint reconstruction. Advanced techniques, such as autoencoders, are used in fingerprint reconstruction to improve images of partially occluded fingerprints. For human activity recognition, the VGG-16 model is used to analyze activities observed in the collected photos and videos, while weapon detection relies on the YOLOv5 and YOLO-NAS architectures for accurate weapon identification. Fingerprint reconstruction is a significant forensic science technique for decoding degraded or incomplete fingerprint data. The technique will improve the quality of fingerprint scans and predict missing features by using advanced algorithms that enable forensic experts to obtain critical information for identification purposes. Fingerprint reconstruction can be a long-trusted, powerful tool for law enforcement.

Piece together fragmented evidence and establish important connections to individuals involved in criminal activities, which is essential for investigations. In weapon detection, modern AI algorithms and computer vision methods are used to automatically identify and classify firearms and other weapons in various situations, thereby ensuring public safety and security. The weapon detection module allows security personnel to respond swiftly to suspicious activities or incidents. Human Activity Recognition (HAR) systems use machine learning techniques to detect a wide range of human behaviors, including typical and atypical ones that could be indicative of potential threats. These capabilities strengthen the overall security measures and allow security teams to quickly identify and respond to security breaches. The outputs from these models are aggregated, processed, and lead to comprehensive conclusions in forensic analysis. Thanks to an intuitive interface that exposes this information, researchers can see and analyze data efficiently. Combining hardware, cloud computing, and advanced deep learning models, the technology simplifies crime scene investigations, improves forensic capabilities, and assists in the prompt and effective administration of justice.



Fig. 2. Two annotated images from the weapon detection dataset [31].

The scene of the crime and crime site investigation captures high-resolution images that go directly to cloud storage and employs machine learning detection in finding guns, blood drops, fingerprints, and human-related activities, all integrated into the reconstruction process to a three-dimensional model.

Data collection

This paper explores the methods used to gather data from different sources following the review of the body of previous research and the plan of action. Getting pictures and recordings of fingerprints, weapons, and suspicious activities is the main goal of data collection for this evaluation. Data is downloaded manually in circumstances where access to the API is limited or nonexistent. For instance, gathered fingerprint data from the FVC 2002 [27] dataset. Created the dataset that was used to train the HAR model by combining the KTH Actions dataset [28] with the Video Fight Detection dataset from Kaggle. The weapon detection-specific dataset [29] uses Roboflow's API. Roboflow is a framework that allows developers to build computer vision models using freely available annotated datasets.

Dataset description

The datasets utilized in this research encompass images of fingerprints, weapons, and videos capturing human activities. Each dataset exhibits distinct characteristics, contributing to the comprehensive analysis conducted in this study. The study used three publicly available fingerprint datasets to ensure that the training inputs were diverse. >300 fingerprint samples are included in the fingerprint dataset. The Kaggle human activity identification dataset [30] contains 10,299 labeled physical activity samples, and the weapon detection dataset [31] contains 3200 annotated photos for concealing and identifying apparent weapons.

Weapon Detection: The weapon identification dataset consists of approximately 2380 annotated images of nine different types of weapons, including firearms, knives, and rifles [31]. The dataset, which is approximately 204 MB in size, contains a variety of YOLO-compatible weapon detection models. The dataset used to train the object detection models comes from Roboflow. There are 2560 JPG-formatted annotated photos in this set of images. Comments are also stored in JSON files. As seen in Fig. 2, every image contains a bounding box around the object and text indicating the object's class. With 14 different weapon classes included in the dataset, weapon detection algorithms may be robustly trained.

Axe, bomb, bow, cleaver, cutlass, katana, knife, mace, machine gun, Morningstar, pistol, rifle, rocket launcher, scabbard, scope, shield, shotgun, sickle, SMG, sniper rifle, spear, sword, and war hammer are the classes. It is divided into the validation, test, and train sets, respectively. A validation collection of 257 photos, a test set of 137 photographs, and a training set of 75 % of 1912 images.

Fingerprint Reconstruction: Among the many fingerprint images available is the FVC2002 dataset, which is used for fingerprint reconstruction jobs. The study's fingerprint collection comprises 110 pictures collected from four different devices. Thirty photographs have been set aside for validation, and eighty pictures have been earmarked for testing. The .tif file format is used for all photos, guaranteeing uniformity and interoperability throughout the collection. This varied dataset allows for a thorough assessment and validation of fingerprint analysis methods and approaches. It includes fingerprint photos collected from different devices, as shown in Fig. 3. **Human Activity Recognition (HAR):** Human Different datasets, including the KTH Actions Dataset and the Video Fight Detection Dataset from Kaggle, are used to train models for human activity detection. Fig. 4 represents the HAR dataset. The human activity dataset includes 10,299 labeled examples of human actions, as well as 5,61 numerical features extracted. This dataset, which was compiled from 30 people walking, sitting, and standing, among other common behaviors, is widely used for sensor-based activity

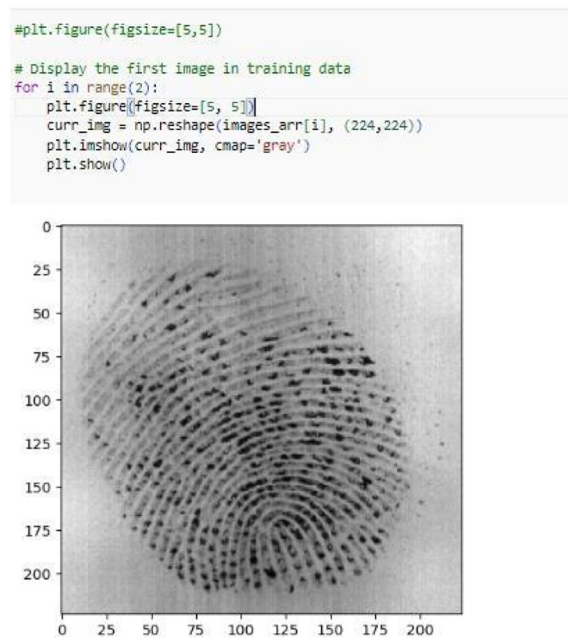


Fig. 3. First image taken from the fingerprint training dataset [32].



Fig. 4. Human Activity Recognition (HAR) dataset [30].

classification. These datasets cover a broad range of human actions, allowing the rich training set of data of HAR algorithms. The data used in the research is in 100 videos for each of three classes: “fights,” “running,” and “walking.” Those video files were kept in the AVI format; therefore, the analysis with its usage to train the models becomes easy. Thus, with three categories, this set of data comprises 300 videos of the HAR dataset. This abundance of data facilitates the confident training and evaluation of HAR algorithms.

Implementation

Various algorithms are presented in the following forensic procedures, all of which are crucial to the effectiveness of the crime prevention strategy. Google Colab and Visual Studio Code were used for execution. A hosted Jupyter Notebook service called Google Colab provides free processing power, including GPUs and TPUs. A source code editor for several types of coding tasks is called VS Code. In addition to a large number of extensions, it offers comprehensive language support.

Autoencoders

One kind of neural network used in unsupervised learning is the autoencoder. Using its lower-dimensional representation, it is utilized to reconstitute original data. Two networks comprise the autoencoder architecture: the encoder network and the decoder network. One or more layers of artificial neurons comprise the encoder network. These layers condense the input data into a latent representation, which is a lower-level representation. It merely keeps track of the most crucial elements. A decoder network receives this compressed representation and uses it to learn how to recover the original input. The autoencoder is designed to minimize the difference between the input and output data by utilizing a loss function during training. Due to their effectiveness in reproducing images while also reducing dimensionality, autoencoders are commonly used for image compression. Their flexibility and adaptability make them valuable in various machine-learning applications. The aim of fingerprint reconstruction is to create images that closely resemble the originals. Typically, this process uses a set of features, such as minutiae, extracted from the original fingerprint as input. Autoencoders are trained on a collection of fingerprints, where the encoder network processes the incomplete image to extract a latent space vector, retaining only the most critical information. The decoder then reconstructs the entire fingerprint image using this latent space representation.

YOLO (You only look once)

Including versions like YOLOv5 and YOLO–NAS, it is a family of deep learning-based object detection algorithms. It works by doing a single pass through the neural network to detect objects in images. The input image is divided into grid cells, and for each cell, bounding boxes and class probabilities are predicted simultaneously. It is particularly effective in real-time applications such as surveillance systems and self-driving cars due to the greatly accelerated inference times. YOLOv5 is a variant of the original YOLO framework, which is known to be efficient with high accuracy on real-time object detection tasks. In its methodology, it divides the input image into grid cells to check for the presence of objects in each cell, and the results are recorded as a vector. After training for 50 epochs to further improve the ability of the model to detect different objects on the labeled dataset from Roboflow named PROJECT-4, containing 14 classes, automated architecture designs with Neural Architecture Search (NAS) [33] approaches have advanced much with the advent of YOLO–NAS to make the YOLO object detection framework improved. Following training for the same number of epochs on an identical dataset, YOLO–NAS takes advantage of its automatic architecture search capabilities to discover the most efficient network architectures for object detection tasks. It explores many possible topologies and then fine-tunes its parameters for both precision and efficiency. By drawing on the same training on the same dataset for the same period, YOLO–NAS demonstrates the ability to produce a highly accurate and effective object detection model. Weapon detection systems, which are capable of identifying and classifying firearms and other weapons on their own from various scenarios using the latest AI algorithms and computer vision techniques, have great potential for enhancing public safety and security. Such techniques are essential in the identification of possible violent crimes during investigations.

VGG16

The VGG-16 architecture is a widely used CNN in computer vision for classification and image recognition tasks. It is comprised of 16 layers, which include convolutional, max-pooling, and fully connected layers. An input image of a given size is passed into the network. Following each convolutional layer in VGG16 are rectified linear units (RELU) activation functions. Each of the layers of the trained filters processes the input image and extracts features. Max-pooling layers are used to reduce spatial dimensions and highlight important characteristics after several convolutional layers. The fully connected layers that follow the final convolutional layers link the extracted features to the output. Softmax activation is used in these layers to generate class probabilities. The output layer produces a vector of probabilities that indicates the likelihood of an input image for each class in particular. The KTH behavior collection was combined with the Video Fight Detection Dataset from Kaggle to result in a varied set of films with different human behaviors. After 20 epochs of training, VGG-16 went through iterative optimization to find and pick up patterns in the combined dataset. This detection and classification in dynamic environments. Human Activity Recognition (HAR) systems play a crucial role in sensing and analyzing the human behavior itself through wearable data, camera feed, etc., especially in cases of security or surveillance. These HAR systems would be able to identify a diverse range of activities, both usual and unusual, which could probably indicate some hidden threats. For this reason, security personnel have the opportunity to identify and, therefore, respond quickly to all security breaches to improve the overall security standards and situational awareness.

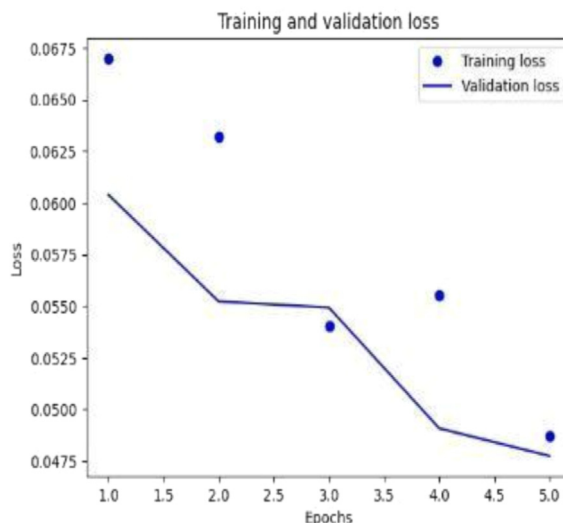


Fig. 5. Training and validation losses.

Table 1

Object Detection Models And Their Respective mAP%.

Algorithm	mAP%
YOLOv5s	47.22
YOLOv5m	57.54
YOLOv5l	63.65
Proposed Model	77

Method validation

The following section represents the results and conclusions that are extracted after training the different models using distinct datasets. These results would represent the benefits of the technology used and are thus helpful for evaluating the generalized performance of the system.

Fingerprint reconstruction

This paper investigates whether autoencoders can be a good tool in reconstructing fingerprints from partially occluded or visible fingerprint data. A dataset of 320 fingerprint images was used to train the model, which has a total of 3146,25 parameters. Fig. 5 plots the training epochs on the x-axis and the corresponding losses on the y-axis. Validation loss (0.0477) and step loss (0.0487) are displayed by the model. Both the training and validation losses exhibit a steady downward trend on the graph. The model's ability to recreate partial fingerprints is demonstrated by the losses' consistent decline. The model was trained using the Adam optimizer with a 0.0001 learning rate. The model runs under 50 epochs with early stopping and validation checks for maintaining performance.

Weapon detection

In this study, Bayesian optimization and grid search were used for hyperparameter tuning to optimize model performance across fingerprint, activity, and weapon detection tasks. A weapon identification dataset with 14 classes to investigate the efficacy of training two object detection models, YOLOv5 and YOLO-NAS submodules (small, medium, and big). The mean average precision (mAP), a commonly used statistic to evaluate the performance of item identification models, as our main assessment metric. At an Intersection over Union (IoU) threshold of 0.5, it is calculated. The mAP takes into account both the accuracy (the number of correct detections) and recall (the number of true positives detected) of the model's predictions. The predicted bounding boxes' and the ground truth bounding boxes' degree of overlap is measured by the IoU threshold.

The performance analysis of a YOLO-NAS model trained on a weapon detection dataset. Consisting of 2560 images. Mean average precision (mAP) increased during 111 epochs on the validation data, reaching a peak of 77.8 %, as seen in Fig. 6(a). The mean absolute percentage at the 50th epoch is 77 %. These results show how effectively the YOLO-NAS model detects weapons in photos, thereby facilitating enhanced security protocols. The graphical representation of our results plots the training epochs of YOLOv5 as displayed in Fig. 6(b) against the corresponding mAP scores on the y-axis. More specifically, Table 1 shows the big model had a mean average

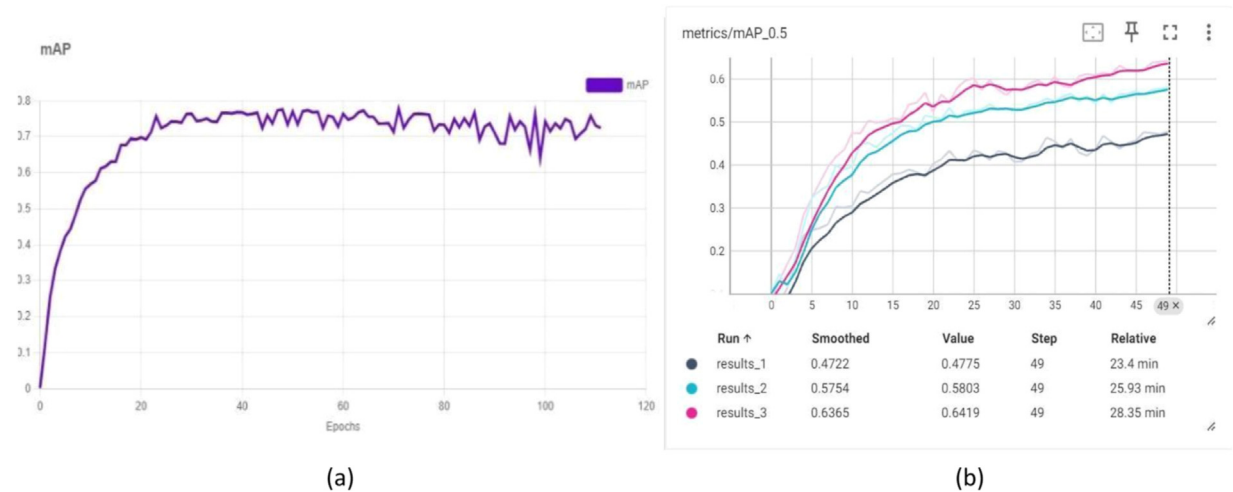


Fig. 6. Line graph showing mAP progression of (a) YOLO-NAS and (b) YOLOv5 submodules.

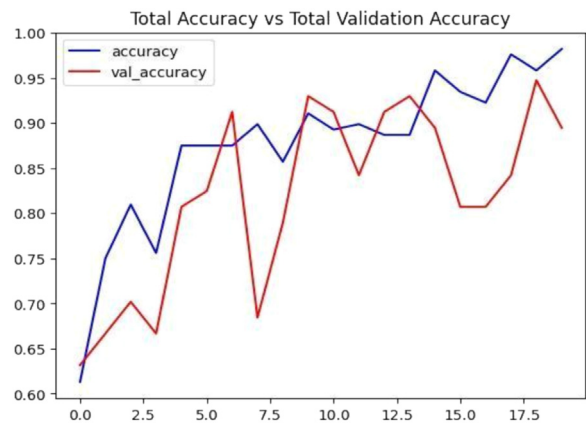


Fig. 7. Total accuracy vs total validation.

precision (mAP) of 0.6365 at the end of training (epoch 50), the medium model had an mAP of 0.5754, and the small model had an mAP of 0.4722. These results indicate a bias toward larger model architectures for weapon identification, even if smaller model architectures train more quickly.

Furthermore, analysis of the training procedure for all 4 models reveals a consistent increase in mAP during the training epochs. With an upward trajectory, it appears that the models are learning positively and using the dataset to enhance their detection skills. When all four are compared, YOLO-NAS performs better and has a higher mAP score; as a result, it will be utilized in the online application to identify firearms and has a higher mAP score; as a result, it will be utilized in the online application to identify firearms.

Human activity recognition

In this work, the Kaggle dataset includes 10,299 labeled sensor readings from 30 individuals performing daily activities, used for human activity recognition. Investigate how well VGG-16 performs when trained on a dataset of 300 movies of various human activities. Fig. 7 illustrates how the model was trained over 20 epochs, yielding an overall accuracy of 98.21 % and a validation accuracy of 89.47 %.

It is 0.0640 overall and 0.2329 for validation. Fig. 8 demonstrates a consistent downward trend in the validation and training losses, suggesting that the model is capable of accurately identifying human behavior.

User interface

By providing authorities with a platform to upload photos or videos from crime scenes and receive analyses like fingerprint reconstruction, weapon detection, and human activity recognition, the user interface (UI) built with Flask aims to expedite forensic

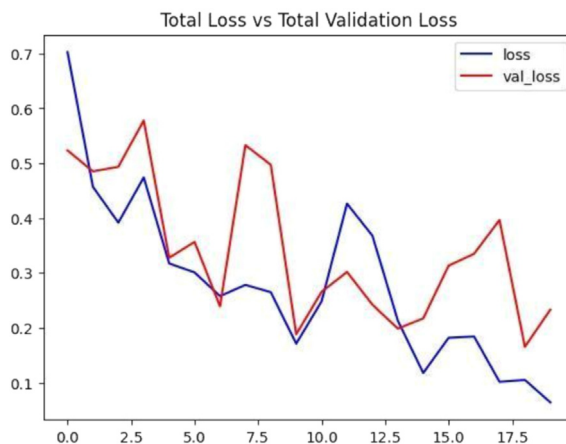


Fig. 8. Total loss vs total validation loss.

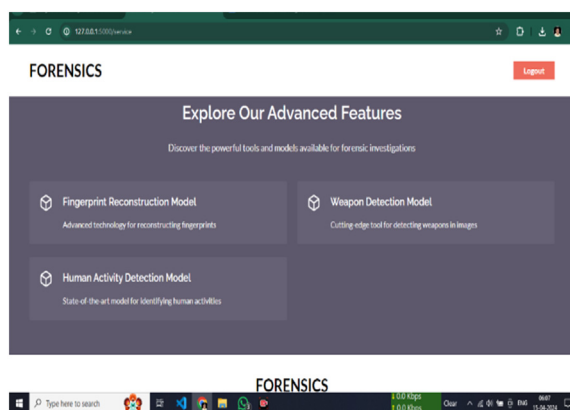


Fig. 9. Explore services page.

investigations. The UI entails several important parts, such as a landing page, registration page, dashboard, login page, explore services, upload page, and output page, for it to be both safe and efficient, as shown in Fig. 9.

Entering the user interface, the interface greets a user with a catchy landing page that states its features and instructions of what to do next. Intuitive options for navigation have made it quite easy for one to move across the registration procedure, which, in turn, ensures accountability and thus access to highly sensitive forensic information. During this process, they are required to provide the person's name, email address, and password as the essential data. This helps in the system's security with regard to their unauthorized access into the system after registration. Using the credentials in the login is possible, opening up the total capabilities of the interface.

Once inside the user interface (UI), users are greeted by an interactive landing page that summarizes the features of the platform and gives a clear call to action for the next steps. Intuitive navigation options allow users to move through the registration process easily, which is critical to ensuring accountability and securing access to sensitive forensic information. Users are required to provide basic information during registration, such as name, email address, and password. All this information is kept safe from unauthorized access to the system. Once registered, users can log in with their credentials to unlock the full capabilities of the interface. Fig. 10. Upload page for fingerprint reconstruction. Users are provided with an intuitive interface, as shown in Fig. 10, that enables users to share images or videos from crime scenes after logging. In order to ensure an interruption-free uploading process, the UI accepts a range of file formats and provides feedback.

The UI launches the desired analysis, such as weapon identification, HAR, and fingerprint reconstruction, after a successful upload. The analysis's findings are shown on the output page. Furthermore, Fig. 11's output page specifies the location of the results' storage for future reference as well as if they have been saved. This feature helps officials monitor the status of analyses and get pertinent data as needed, which guarantees accountability and transparency in forensic investigations.

The uploaded image is displayed on the interface in Fig. 12(a). The image that was shared shows a synthetic fingerprint minutia, or the ridges on fingers that allow for fingerprint identification. The contents of the "reconstructed_image_iter_None.png" output file are displayed in Fig. 12(b). It is a reconstruction of the fingerprint image that was uploaded.

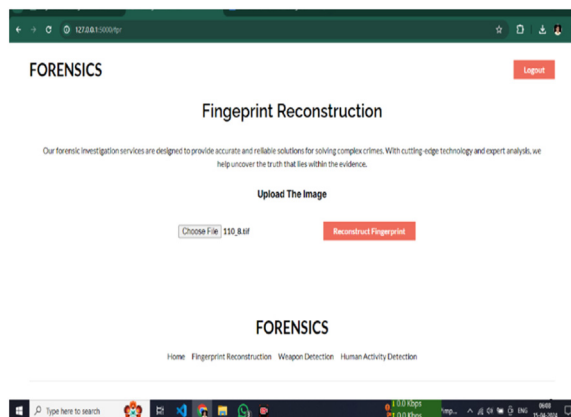


Fig. 10. Upload page for fingerprint reconstruction.

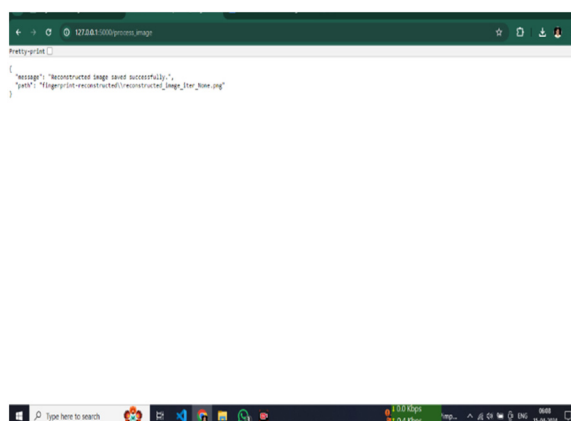


Fig. 11. Output page for fingerprint reconstruction.

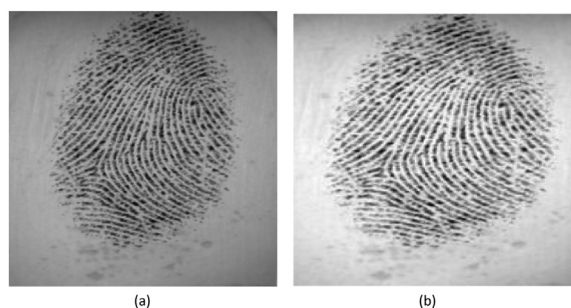


Fig. 12. Results showing (a) Uploaded image and (b) the output image.

Conclusion

This study suggests a modern technique of crime prevention that blends intelligent models, cloud-based data processing, and technology. Our proposed web application combines three distinct functionalities within a single platform, introducing newness to current literature. The system uses advanced equipment with cameras to collect photographs and videos, providing a thorough view of the crime scene. Cloud-based technologies and intelligent model inference provide high-performance weapon detection, activity recognition, and fingerprint reconstruction. Autoencoders can correctly recreate fingerprints with a validation loss of 0.0477 and a step loss of 0.0487. YOLO-NAS outperforms the four modes taught to detect objects, with a mAP of 77.8 %. VGG16, utilized for HAR, has a total accuracy of 98.21 %. The user-friendly interface and robust output enable authorities to access and interpret data more readily, resulting in better decision-making. Moving on, the project may incorporate IoT technology to enable real-time support. The development of an end-to-end gadget capable of capturing and analyzing real-time or near-real-time crime scenes.

Limitations

None.

Ethics statements

None.

CRediT author statement

Vanita Kshirsagar: Writing, Conceptualization, Methodology, Software, Writing- Original draft preparation, Validity tests, **Nishant Pachpor:** Data identification, Visualization, **Shubhangi Suryawanshi:** Reviewing. **Tanvi Chavan:**Supervision. **Navya J. Nair:**Validation, **Purvesh Agrawal:** Editing, **Tejas Shahane:** Investigation.

Supplementary material *and/or* additional information [OPTIONAL]

None.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

I have downloaded the data from www.kaggle.com.

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None.

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